

34) Suppose that g is a function from A to B and f is a function from B to C . Prove each of these statements.

a) If $f \circ g$ is onto, then f must also be onto.

Solution: Given, $f \circ g$ is onto. By Definition 7, $\forall z \in C, \exists y \in A$ s.t. $f(g(y)) = z$. By defn, $g(y) \in B$.

$\therefore \forall z \in C, \exists z \in B$ s.t. $f(z) = z$ and $z = g(y)$ for some $y \in A$.

By Defn. 7, f is onto. $\square$

b) If $f \circ g$ is one-to-one, then g must also be one-to-one.

Solution: By Defn. 5, if $x, y \in A$ and $f(g(x)) = f(g(y))$, then $x = y$.

$\therefore x = y$, now $g(x) = g(y) \in B$ by defn.

By Defn. 5, g is one-to-one. $\square$

c) If $f \circ g$ is a bijection, then g is onto if and only if f is one-to-one.

Solution: By Defn. 8, $f \circ g$ is both one-to-one and onto.

By Defn. 5, if $x, y \in A$ and $f(g(x)) = f(g(y))$, then $x = y$.

By Defn. 7, $\forall z \in C, \exists y \in A$ s.t. $f(g(y)) = z$.

Let g be onto. By Defn. 7, $\forall z \in B, \exists x \in A$ s.t. $g(x) = z$.

By part (b) if $x, y \in A$ and $f(g(x)) = f(g(y))$, then $g(x) = g(y)$.

Let $z, w \in B$ and $f(z) = f(w)$.

By Defn. 7, $\exists x, y \in A$ s.t. $z = g(x)$ and $w = g(y)$.

$\therefore f(g(x)) = f(g(y))$ By part (b), $g(x) = g(y)$. $\therefore z = w$.

By Defn. 5, f is one-to-one.

Let f be one-to-one. By Defn. 5, if $x, y \in B$ and $f(x) = f(y)$, then $x = y$.

By part (a), f is onto. By Defn. 7, $\forall z \in C, \exists w \in B$ s.t. $f(w) = z$.

We know, $\forall z \in C, \exists y \in A$ s.t. $f(g(y)) = z$.

~~Let g not be onto. $\therefore \exists z \in B$ s.t. $\forall x \in A, g(x) \neq z$.~~